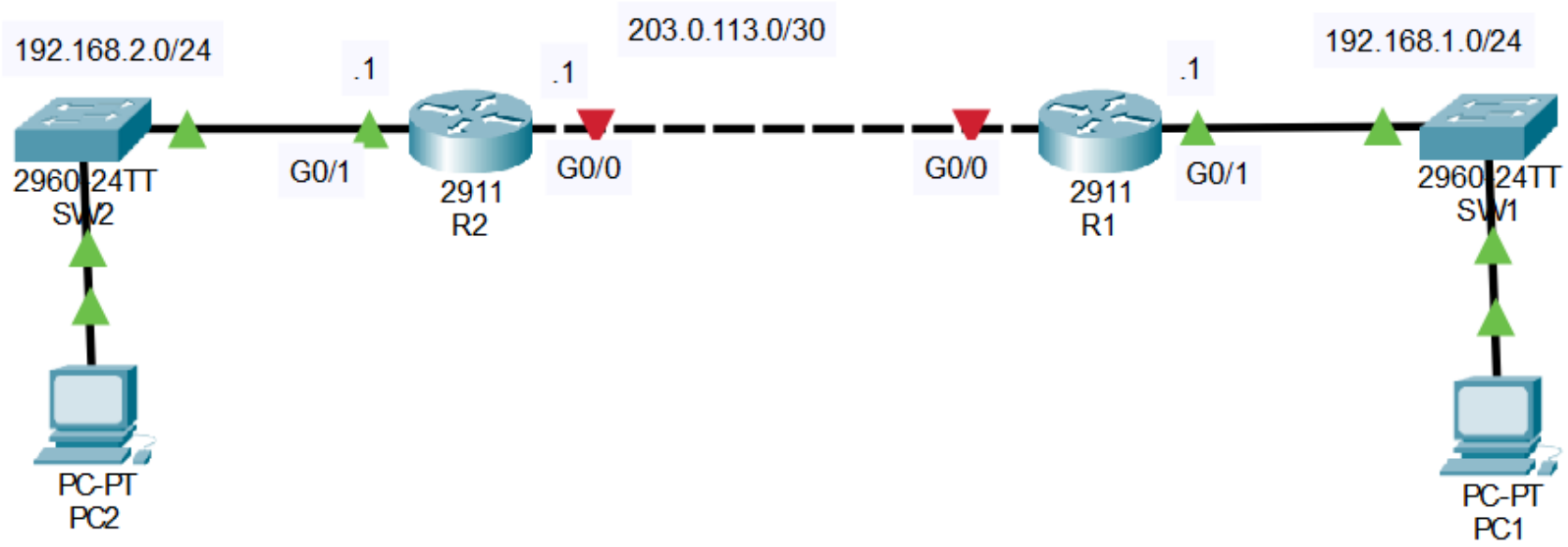
Logical | Physical x: 225, y: 1



1. Configure the follow ing DHCP pools on R2:
POOL1: 192.168.1.0/24 (reserve .1 to .10)
    DNS 8.8.8.8
    Domain: AnojLab.com
    Default Gatew ay: R1
POOL2: 192.168.2.0/24 (reserve .1 to .10)
    DNS 8.8.8.8
    Domain: AnojLab.com
    Default Gatew ay: R2
POOL3: 203.0.113.0/30 (reserve .1)

2. Configure R1's G0/0 interface as a DHCP client.
   What IP address did it configure?

3. Configure R1 as a DHCP relay agent for the 192.168.1.0/24 subnet.

4. Use the CLI of PC1 and PC2 to make them request an IP address
   from their DHCP server.

R2

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up


R2>
R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.10
R2(config)#ip dhcp excluded-address 192.168.2.1 192.168.2.10
R2(config)#dns-server 8.8.8.8
                ^
% Invalid input detected at '^' marker.

R2(config)#ip dhcp excluded-address 203.0.113.1
R2(config)#ip dhcp pool POOL1
R2(dhcp-config)#network 192.168.1.0 255.255.255.0
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#
R2(dhcp-config)#domain-name AnojLab.com
R2(dhcp-config)#default-router 192.168.1.1
R2(dhcp-config)#
```

Copy Paste

Top

```
R2(dhcp-config)#default-router 192.168.1.1
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#ip dhcp pool POOL2
R2(dhcp-config)#network 192.168.2.0 255.255.255.0
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#domain-name AnojLab.com
R2(dhcp-config)#default-router 192.168.2.1
R2(dhcp-config)#
R2(dhcp-config)#
```

Co

```
R2(dhcp-config)#default-router 192.168.2.1
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#
R2(dhcp-config)#ip dhcp pool POOL3
R2(dhcp-config)#network 203.0.113.0 255.255.255.252
R2(dhcp-config)#
R2(dhcp-config)#
```

```
R2(dhcp-config)#network 203.0.113.0 255.255.255.252
R2(dhcp-config)#
R2(dhcp-config)#do sh run | section dhcp
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 203.0.113.1
ip dhcp pool POOL1
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 8.8.8.8
 domain-name AnojLab.com
ip dhcp pool POOL2
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 8.8.8.8
 domain-name AnojLab.com
ip dhcp pool POOL3
 network 203.0.113.0 255.255.255.252
R2(dhcp-config)#
```

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R1

```
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!



R1>
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

R1(config-if)#
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0 assigned DHCP address
203.0.113.2, mask 255.255.255.252, hostname R1


R1(config-if)#
```

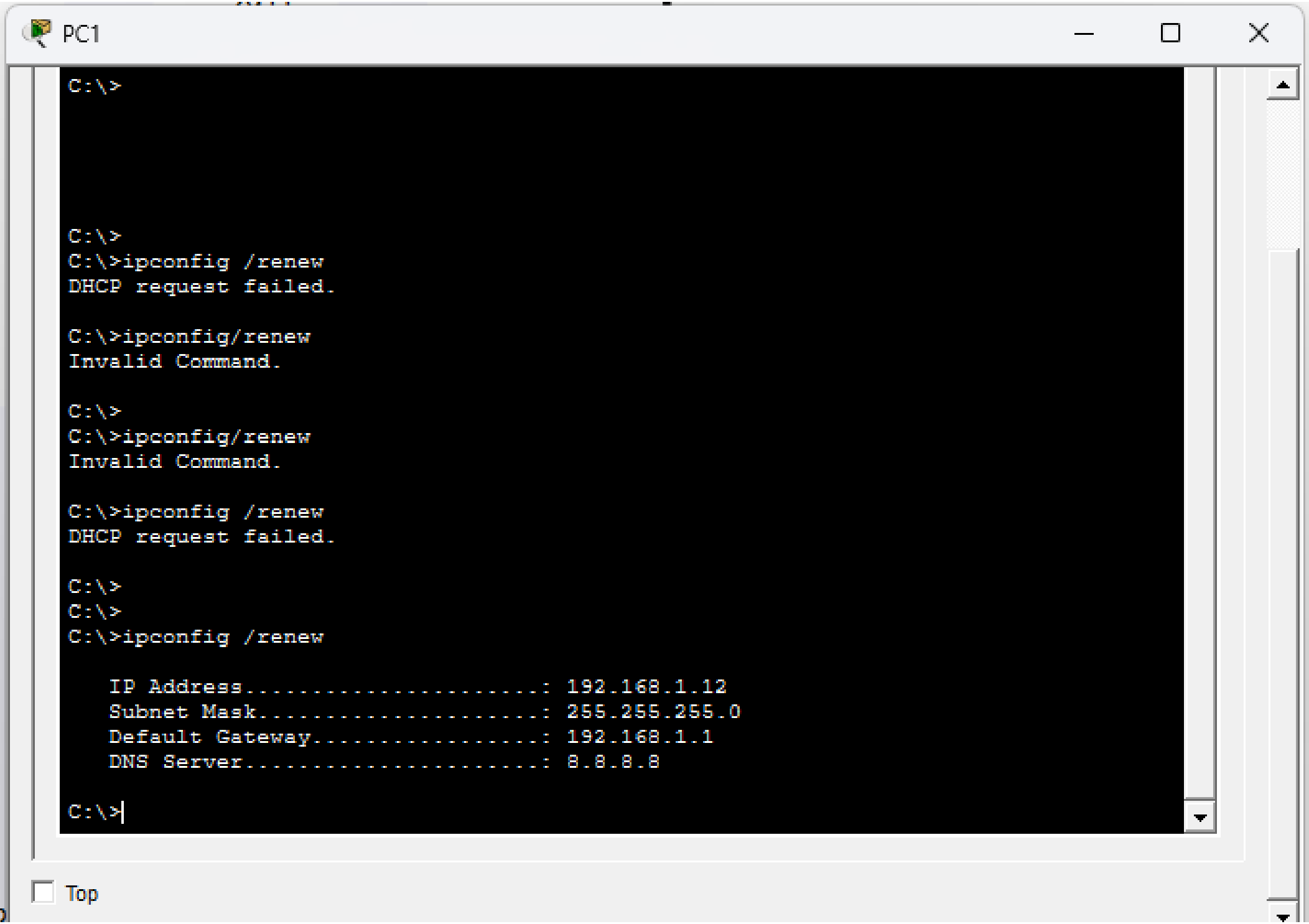
Copy Paste

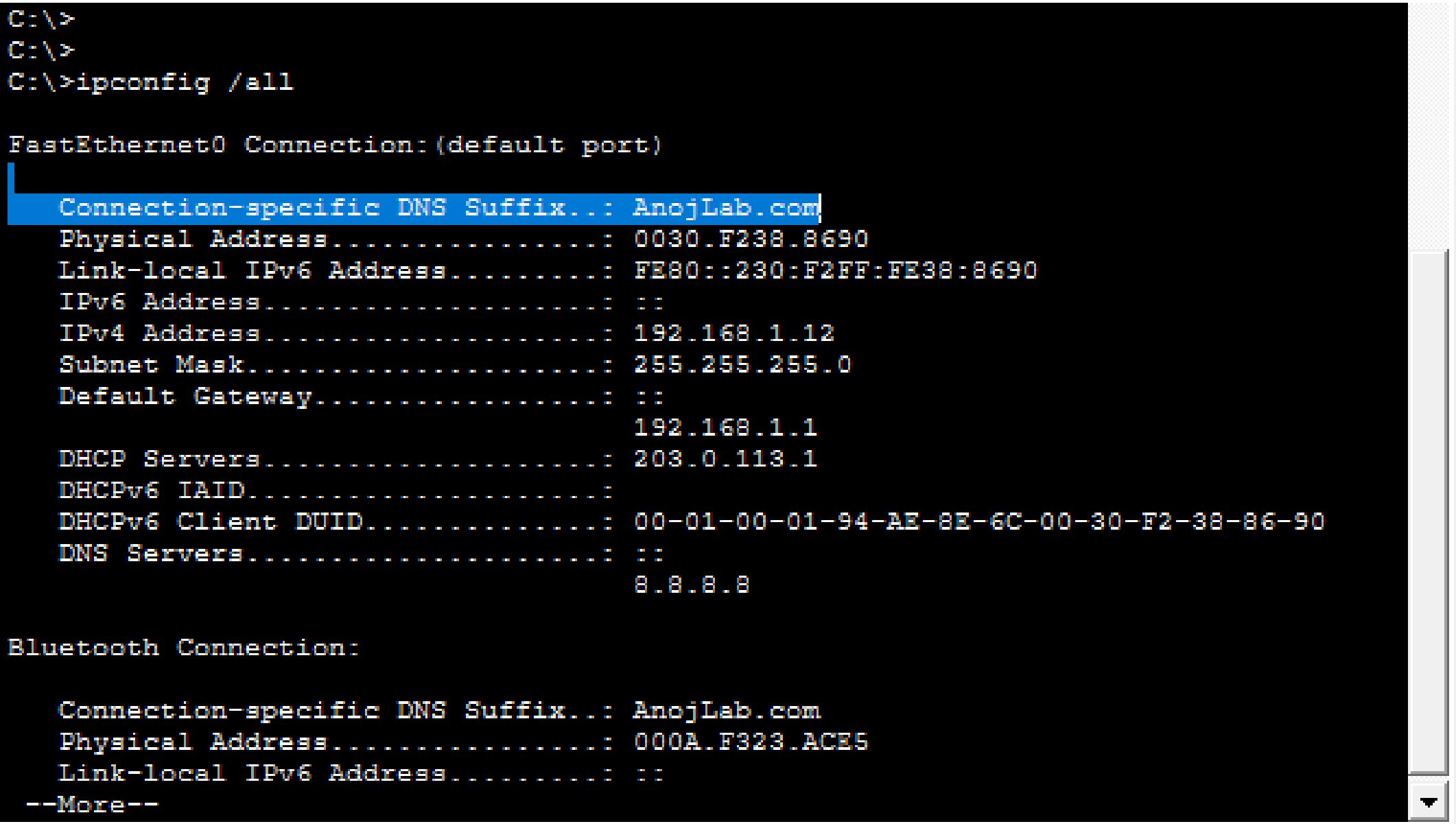
Top

```
203.0.113.2, mask 255.255.255.254, hostname R1


R1(config-if)#
R1(config-if)#
R1(config-if)#int g0/1
R1(config-if)#ip helper-address 203.0.113.1
R1(config-if)#
R1(config-if)#
```

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PC1
C:\>
C:\>
C:\>ipconfig /renew
DHCP request failed.
C:\>ipconfig/renew
Invalid Command.
C:\>
C:\>ipconfig/renew
Invalid Command.
C:\>ipconfig /renew
DHCP request failed.
C:\>
C:\>
C:\>ipconfig /renew
IP Address.....: 192.168.1.12
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.1.1
DNS Server.....: 8.8.8.8
C:\>
Top


C:\>
C:\>
C:\>ipconfig /all
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..: AnojLab.com
Physical Address.....: 0030.F238.8690
Link-local IPv6 Address.....: FE80::230:F2FF:FE38:8690
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.12
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.1.1
DHCP Servers.....: 203.0.113.1
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-94-AE-8E-6C-00-30-F2-38-86-90
DNS Servers.....: ::
8.8.8.8
Bluetooth Connection:
Connection-specific DNS Suffix..: AnojLab.com
Physical Address.....: 000A.F323.ACE5
Link-local IPv6 Address.....: ::
--More--

Top